

Ammonia 4-Stroke Gas Engine Study

Joint YTL × MYSGH2 timeline — maturity, carbon emissions & performance-contract assessment across the shortlisted OEM set | Kickoff: **Tue 25 Aug 2026** → Finish: **Tue 22 Sep 2026** (4 calendar weeks / 20 working days)

This document details the work timeline agreed between YTL (IPP, Singapore) and MYSGH2 to study the technology risk (maturity + carbon-emission performance) and contractual (performance-guarantee) structure of 4-stroke ammonia gas engines across the shortlisted OEM set. Direct OEM engagement is led by MISC — the partner company under the parent group that holds the existing OEM relationships — and runs as a single continuous item rather than a sequence of MYSGH2-run RFI/discussion rounds. This document expands the milestone plan into day-level tasks with owners and dependencies, mirroring the accompanying Gantt-chart workbook (whose Gantt Chart tab is formula-linked to its Task List tab, so edits to one drive the other).

1. Milestone Summary

Date	Milestone	Description
Fri 21 Aug 2026	Prep complete	OEM shortlist (confirmed with MISC), kickoff charter, data-requirement brief for MISC, and NDA status all confirmed ahead of kickoff.
Tue 25 Aug 2026	KICKOFF MEETING	Formal study kickoff with YTL: objectives, scope, timeline and success criteria aligned.
Fri 4 Sep 2026	Week 2 status review	Progress check-in with YTL: MISC's OEM engagement progress and Technology Risk Assessment framework v1 status.
Fri 11 Sep 2026	Week 3 status review	Progress check-in with YTL: Technology Risk Assessment scorecard (maturity + carbon) populated.
Mon 21 Sep 2026	Draft report issued	Consolidated draft report (Technology Risk Assessment, performance-contract framework, risk register) submitted for YTL review.
Tue 22 Sep 2026	FINAL REPORT & CLOSE-OUT	YTL comments incorporated; final report delivered; close-out session presenting findings and recommended next steps.

2. Detailed Task Timeline

Full task breakdown by phase, with owner, working-day duration and predecessor(s). Milestones are shaded gold; task rows are shaded by workstream (see legend, §4).

Phase	ID	Task	Owner	Start	Finish	Dur. (wd)	Predecessor(s)
Prep	P1	Finalize OEM shortlist & confirm with MISC	MYSGH2/MISC	Mon 17-Aug	Tue 18-Aug	2	-
Prep	P2	Draft kickoff agenda & study charter	MYSGH2	Mon 17-Aug	Wed 19-Aug	3	-
Prep	P3	Align data/evidence requirements to hand to MISC	MYSGH2	Tue 18-Aug	Thu 20-Aug	3	P1
Prep	P4	Confirm NDA status per OEM (via MISC)	MISC	Wed 19-Aug	Fri 21-Aug	3	P1

Phase	ID	Task	Owner	Start	Finish	Dur. (wd)	Predecessor(s)
Prep	P5	Internal YTL/MYSGH2/MISC alignment call	Both	Fri 21-Aug	Fri 21-Aug	1	P2, P3, P4
Week 1	M1	KICKOFF MEETING	Both	Tue 25-Aug	Tue 25-Aug	1	P5
Week 1	W1.1	Agree Technology Risk Assessment framework skeleton	MYSGH2	Wed 26-Aug	Thu 27-Aug	2	M1
Week 1	W1.2	Finalize workstream owners & roles	Both	Mon 31-Aug	Mon 31-Aug	1	M1
Week 1	OEM1	OEM ENGAGEMENT & DATA COLLECTION (led by MISC)	MISC	Mon 31-Aug	Wed 16-Sep	13	M1, P4
Week 2	W2.1	Build Technology Risk Assessment framework v1 (maturity TRL/CRI scale + carbon-emission methodology)	MYSGH2	Tue 01-Sep	Mon 07-Sep	5	W1.1
Week 2	W2.2	Weekly status check-in (Wk2)	Both	Fri 04-Sep	Fri 04-Sep	1	OEM1
Week 3	W3.1	Populate Technology Risk Assessment scorecard (maturity + carbon, per OEM)	MYSGH2	Tue 08-Sep	Fri 11-Sep	4	W2.1, OEM1
Week 3	W3.2	Draft performance-contract framework	MYSGH2/YTL	Thu 10-Sep	Mon 14-Sep	3	OEM1
Week 3	W3.3	Weekly status check-in (Wk3)	Both	Fri 11-Sep	Fri 11-Sep	1	W3.1
Week 4	W4.1	Build risk & derisking framework (fuel-supply, safety/toxicity, regulatory, counterparty)	MYSGH2	Tue 15-Sep	Thu 17-Sep	3	W3.1, W3.2
Week 4	W4.2	Consolidate all findings	MYSGH2	Thu 17-Sep	Fri 18-Sep	2	W4.1, OEM1
Week 4	W4.3	Draft consolidated report	MYSGH2	Fri 18-Sep	Mon 21-Sep	2	W4.2
Week 4	W4.4	Internal YTL review of draft report	YTL	Mon 21-Sep	Mon 21-Sep	1	W4.3
Finish	F1	Incorporate YTL review comments	MYSGH2	Tue 22-Sep	Tue 22-Sep	1	W4.4
Finish	F2	FINAL REPORT & CLOSE-OUT SESSION	Both	Tue 22-Sep	Tue 22-Sep	1	F1

3. Workstream × Week View

Workstream	Prep (17-21 Aug)	Wk1 (25-31 Aug)	Wk2 (1-7 Sep)	Wk3 (8-14 Sep)	Wk4 (15-21 Sep)	Finish (22 Sep)
Governance	Alignment call	Kickoff	Status check-in	Status check-in	Internal review	Close-out
OEM Engagement (MISC-led)	NDAs confirmed, data-requirements handed to MISC	Engagement starts (31 Aug)	Ongoing (single continuous item through 16 Sep)	Ongoing (single continuous item through 16 Sep)	Completes 16 Sep	-
Technology Risk Assessment	-	Framework skeleton	Framework v1 (maturity + carbon)	Scorecard populated (maturity + carbon)	Consolidated	Finalized
Contract	-	-	-	Framework v1 (guarantees, LDs)	Aligned w/ commercial	Finalized
Risk	-	-	-	-	Register + mitigations	Finalized
Reporting	-	-	-	-	Draft report	Final report

4. Legend

	Category	Description
	Milestone	Decision point, checkpoint, or deliverable hand-off
	Governance	Kickoff, status check-ins, internal YTL/MYSGH2/MISC alignment & sign-off
	OEM Engagement (MISC-led)	MISC (parent-company partner with the existing OEM contacts) owns all direct OEM engagement as ONE continuous item — introductions, data collection, follow-ups & Q&A — not separate MYSGH2-run RFI or discussion-round tasks
	Technology Risk Assessment	Merged maturity (TRL/CRI-style scale) + carbon-emission methodology (incl. N2O/GWP273, NOx, NH3 slip) assessment and per-OEM scorecard
	Contract	Performance-contract framework: availability, efficiency, emissions guarantees, LDs, warranty
	Risk	Risk & derisking framework: fuel-supply, safety/toxicity, regulatory, counterparty risk (technology risk lives in Technology Risk Assessment above)
	Reporting	Findings consolidation, draft & final report production

5. Notes & Assumptions (per CLAUDE.md §7 No-Fabrication Rule)

- This is a **proposed schedule**, not a sourced technical claim — every OEM-facing date is provisional until confirmed by MISC.

- **Decision:** OEM engagement is run entirely by **MISC** — our partner company under the parent group, which holds the existing OEM relationships — not by MYSGH2 or YTL directly. It is modelled as **one continuous item/timeline** ("OEM Engagement & Data Collection", 31 Aug–16 Sep) covering introductions, data collection, follow-ups and Q&A, rather than separate RFI/discussion-round tasks. The dates shown are MYSGH2's planning assumption for when MISC's outreach needs to complete to feed the Technology Risk Assessment — confirm actual cadence with MISC.
- **Decision:** Maturity assessment and carbon-emission assessment are merged into a single **Technology Risk Assessment** workstream (framework build → per-OEM scorecard), rather than run as two separate workstreams.
- **Assumption:** "Performance contract" is scoped as the **framework** for guarantee structure (availability, efficiency/heat-rate, emissions, warranty, liquidated damages) — not a negotiated term sheet with any specific OEM. Confirm with YTL if the latter is wanted by 22 Sep, as that widens scope.
- **Assumption:** The Technology Risk Assessment's maturity component uses a TRL-style (or CRI-equivalent) scale populated from OEM-supplied evidence (test hours, reference installations, certification status) relayed via MISC, cross-checked against the sourced OEM data already held in the MYSGH2 project memory (Wartsila 25 Ammonia, MAN AmmoniaMot 2, Baker Hughes/Hanwha NovaLT16, IHI IM270, Mitsubishi Power H-25).
- Its carbon component must explicitly include **N₂O** (GWP ≈273×CO₂, 100-yr) and **NH₃ slip**, not just CO₂/NO_x — the ammonia-specific risk generic combustion-emissions templates typically miss.
- Four calendar weeks is **tight** for MISC to run OEM engagement across a full shortlist in parallel with the Technology Risk Assessment build. If the OEM list is long (5+), flag a scope/timeline trade-off to YTL and MISC at kickoff.
- Working-day durations exclude weekends; the accompanying Gantt-chart workbook (YTL_Ammonia_Gas_Engine_Study_Gantt.xlsx) computes duration via NETWORKDAYS(), and its Gantt Chart tab is formula-linked to its Task List tab — editing a task's dates, workstream, type or owner in Task List updates the Gantt bars and colors automatically.

Source: MYSGH2 project timeline, drafted 2026-08-14, anchored to the agreed kickoff (25 Aug 2026) and finish (22 Sep 2026) dates. Companion files: power_generation/YTL_ammonia_gas_engine_OEM_study_timeline.md, YTL_Ammonia_Gas_Engine_Study_Gantt.xlsx.